

Section 4.7: Comparing Bluetooth and Wi-Fi

Wi-Fi and Bluetooth are the two main ways that computers can **communicate** with each other **wirelessly**.

In this section we will discuss how each of these work as well as comparing them with each other.

Key Concepts of this section:

- # Understand that both Wi-Fi and Bluetooth allow for the creation of **wireless** networks.
- # Know how computers can use Wi-Fi and Bluetooth to **connect** to a network.
- # Be able to **contrast** and **compare** Wi-Fi and Bluetooth.

Bluetooth and Wi-Fi

- # The two main technologies that computers can use to **communicate wirelessly** are listed below:

- [Wi-Fi](#)
- [Bluetooth](#).

Remember:

Wirelessly means to connect computers or devices together **without** using **wires** or **cables**.

We will look at how computers connect to networks using both bluetooth and Wi-Fi and compare the two.

1 Wi-Fi

What is Wi-Fi?

- # Wi-Fi allows computers/devices to **connect** to each other **wirelessly**, without using cables or wires.

The computers communicate with each other by using **signals** that are broadcast using **radio waves**.

- # In order to use Wi-Fi, you need to have access to the **internet** and a **router**.

Areas where computers/devices can access the internet using Wi-Fi are known as **hotspots**.

- # **Routers** are commonly used to provide a **hotspot** for other computers to connect to the internet.

The router will be connected to the internet through an **ISP (Internet Service Provider)** and will transmit a wireless signal that wireless devices **nearby** can tap into.

This allows **multiple** devices to share the same internet connection.

#

Routers can only broadcast the wireless signal over a **limited** distance. If a device is out of range, it will not be able to tap into the signal.

The range limit of a typical router is between **50m** and **100m**.

How do computers connect to a Wi-Fi network?

- # In order to connect to a network using Wi-Fi you would need:
 - A **computer** or **device**
 - A **wireless network interface card** (Wi-Fi is built-in with many modern devices)
 - Access to a **wireless hotspot** (usually through a router).



Key Words:

Wi-Fi, Bluetooth.

Examples:



Wi-Fi and Bluetooth are the two main ways to connect to networks wirelessly.

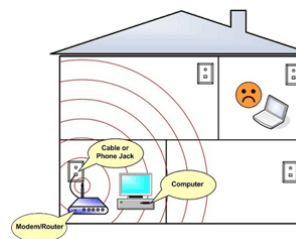
Examples:



Wi-Fi allows computers to communicate using radio waves.



Wi-Fi hotspots allow devices to share the same internet connection.



Wi-Fi signals have a range limit.

2 Bluetooth

What is Bluetooth?

- # Bluetooth is also known as a **Wireless Personal Area Network (WPAN)**.
- # Bluetooth is a form of **wireless** technology that uses **radio waves** to transfer data over distances of between **1 and 100 metres**.

Devices that have this technology contain one of **three classes** of Bluetooth:

- **Class 1** - Most powerful - can transmit data up to **100m**
- **Class 2** - Most common - can transmit data up to **10m**
- **Class 3** - Least powerful - can transmit data up to **1m**.

How do computers connect to a Bluetooth network?

- # In order to connect to a network using Bluetooth you would:
 - Need a **computer** or **device** with **bluetooth** technology
 - Turn bluetooth **on** using your device's settings
 - Let your device **scan** for other bluetooth devices in **range**
 - Send a request to the other bluetooth to **'pair'** (join) with it
 - Once paired, you could **share data** between the devices (**while in range**).

Uses of Bluetooth:

- # Bluetooth has a wide-range of **uses**. some of these are summarised in the table below:

Use input/output devices wirelessly

➔ Devices like **keyboards, mice, printers** and **speakers** can be connected to a computer system without the need for messy wires.
(This can help prevent physical hazards like tripping and helps keep your workspace neat)

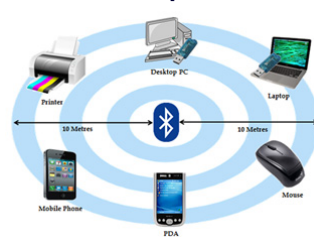
Sharing mobile phone data between devices

➔ Data like **contact numbers** can easily be transferred from one phone to another.
(This is perfect if you get a new mobile phone and don't want to re-enter your contact information)

Connecting to hands-free headsets

➔ **Wireless headsets** allow people to talk into their mobile phones whilst driving.
(This frees up their hands and allows them to concentrate on driving - much safer)

Examples:



Bluetooth networks are known as 'personal networks' because devices have to be close together in order to connect. (click to zoom)



When Bluetooth is switched on, your device will search for other Bluetooth devices to connect to.



Bluetooth hand-free headsets make driving much safer

Comparing Wi-Fi and Bluetooth

- # The table below compares and contrasts the **similarities** and **differences** between Wi-Fi and Bluetooth:

Comparison	Wi-Fi	Bluetooth
Data speeds	Data transfer speeds of up to 250Mbps . (Equivalent to downloading about six mp3 music tracks a second)	Data transfer speeds of up to 25Mbps . (Equivalent to downloading less than one mp3 music track a second)
Winner = Wi-Fi		
Signal range	Range limit of up to 100m .	Most Bluetooth devices have a range limit of up to 10m .
Winner = Wi-Fi		
Security	256-bit encryption used to protect transmitted data.	128-bit encryption used to protect transmitted data.
Winner = Wi-Fi		
Power usage	Uses more power.	Uses less power.
Winner = Bluetooth		
Signal reliability	Wi-Fi signal can be interrupted by electrical equipment.	Bluetooth signal is not as prone to interruption.
Winner = Bluetooth		
Max number of connected devices	Up to 255 devices per router.	Up to 7 devices.
Winner = Wi-Fi		

Activity!

[Differences between Wi-Fi and Bluetooth - Research Sheet](#)

Click the above task and answer the questions about the differences between Wi-Fi and Bluetooth.

Examples:



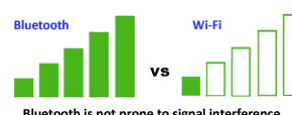
Wi-Fi can transfer data much faster than Bluetooth.



Data sent over Wi-Fi is more secure than data sent over Bluetooth.



Bluetooth won't drain your device's battery as quickly as Wi-Fi will.



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